INCH-POUND

MIL-M-28787/541(NW)
AMENDMENT 1
3 August 1994

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

HIGH PERFORMANCE EMULATION CONTROL II MODULE, KEY CODE FWA

This amendment forms a part of Military Specification MIL-M-28787/541, dated 5 May 1989, and is approved for use by the Crane Division, Naval Surface Warfare Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4 THROUGH 11

TABLE I: Delete and substitute new table I as printed on pages 3 through 10 of this amendment.

PAGE 23

6.3.1, Input parameters, Vic min (V) column, Signals "CIII INTR EN" and "MPIOMPC VINT": Delete "-1.5" (two places) and substitute "NA" (two places).

PAGE 26

6.3.1, Input parameters, Vic min (V) column, Signals "DEST BUS" and "UIR 5": Delete "-1.5" (two places) and substitute "NA" (two places).

PAGES 28 AND 29

6.3.2, Output parameters, Ios max (mA) column: Delete Ios max (mA) column.

PAGE 29

6.3.2, Output parameters, Iozh max (mA) and Iozl max (mA) columns, Signal "JMP CON TRUE": Delete "0.05" (two places) and substitute "0.10" (two places).

AMSC N/A FSC 5963
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PAGES 58 THROUGH 61

TABLE VI: Delete and substitute new table VI as printed on pages 11 through 13 of this amendment.

PAGES 82 AND 83

TABLE XI: Delete and substitute new table XI as printed on pages 14 and 15 of this amendment.

PAGES 84 THROUGH 87

TABLE XII: Delete and substitute new table XII as printed on pages 16 through 19 of this amendment.

PAGE 113

TABLE XVIII, Test condition R56, Max limit: Delete "75" and substitute "100".

Preparing activity:
Navy - NW

(Project 5963-N864)

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TABLE I. Module electrical requirements.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	-------Module shall pass all remaining tests-----						
D1	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D2	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	mA
D3	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D4	Ibd	V	S	-0 02	0.12	-0.02	0.12	-0 02	0.12	mA
D5	Ibd	V	S	-0.02	1.02	-0.02	1.02	-0.02	1.02	mA
D6,D7	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D8	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D9	Ibd	V	S	-0.02	1.02	-0.02	1.02	-0.02	1.02	mA
D10	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D11	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
D12	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D14	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D15	Ibd	V	S	0.0	2.2	0.0	2.2	0.0	2.2	mA
D16	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D17-D23	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D24	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D25	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D26	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D27-D32	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D33	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D34	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
D35,D36	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D37-D39	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
D40	Ibd	V	S	-0.02	1.12	-0.02	1.12	-0.02	1.12	mA
D41,D42	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D43	Ibd	V	S	-0.02	1.02	-0.02	1.02	-0.02	1.02	mA
D44,D45	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D46	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D47	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D48	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D49	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D50	Ibd	V	S	0.0	1.3	0.0	1.3	0.0	1.3	mA
D51	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D52	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D53-D55	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D56	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D57-D62	Ibd	V	S	-0.02	0.12	-0.02	0.12	-0.02	0.12	mA
E1-E28	Ios	VI	S	------Module shall pass all remaining tests------						
F1	Ilh	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F2	Ilh	VII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA
F3	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F4	Ilh	VII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
F5	Ilh	VII	X	-3.06	-0.51	-3.06	-0.51	-3.06	-0.51	mA
F6	Ilh	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F7	Ilh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F8	Ilh	VII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
F9	Ilh	VII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
F10	Ilh	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F11	Ilh	VII	X	-1.7	0.09	-1.7	0.09	-1.7	0.09	mA
F12	Ilh	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F13	Ilh	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F14	Ilh	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F15	Ilh	VII	X	-5.0	0.14	-5.0	0.14	-5.0	0.14	mA
F16	Ilh	VII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA

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Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F17-F19	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F20	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F21,F22	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F23	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F24	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F25	Iih	VII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
F26	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F27,F28	Iih	VII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
F29-F32	Iih	VII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
F33	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F34	Iih	VII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
F35,F36	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F37-F39	Iih	VII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
F40	Iih	VII	X	-3.26	-0.49	-3.26	-0.49	-3.26	-0.49	mA
F41	Iih	VII	X	-1.68	0.08	-1.68	0.08	-1.68	0.08	mA
F42	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F43	Iih	VII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
F44,F45	Iih	VII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
F46	Iih	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F47	Iih	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F48	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F49	Iih	VII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
F50	Iih	VII	X	-2.05	0.09	-2.05	0.09	-2.05	0.08	mA
F51	Iih	VII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
F52	Iih	VII	X	-2.28	0.1	-2.28	0.1	-2.28	0.1	mA
F53,F54	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F55	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F56	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F57-F59	Iih	VII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
F60-F62	Iih	VII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G1	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G2	Iil	VIII	X	-1.1	0.07	-1.1	0.07	-1.1	0.07	mA

- See footnotes at end of table.

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Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G3	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4	III	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G5	III	VIII	X	-3.06	-0.51	-3.06	-0.51	-3.06	-0.51	mA
G6	III	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G7	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G8	III	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
G9	III	VIII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
G10	III	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G11	III	VIII	X	-1.7	0.09	-1.7	0.09	-1.7	0.09	mA
G12	III	VIII	X	-0.25	0.02	-0.25	0.02	-0.25	0.02	mA
G13	III	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G14	III	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G15	III	VIII	X	-5.0	0.14	-5.0	0.14	-5.0	0.14	mA
G16	III	VIII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
G17-G19	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G20	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G21,G22	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G23	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G24	III	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G25	III	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G26	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G27,G28	III	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G29-G32	III	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G33	III	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G34	III	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
G35,G36	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G37-G39	III	VIII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
G40	III	VIII	X	-3.26	-0.49	-3.26	-0.49	-3.26	-0.49	mA
G41	III	VIII	X	-1.68	0.08	-1.68	0.08	-1.68	0.08	mA
G42	III	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G43	III	VIII	X	-1.31	-0.54	-1.31	-0.54	-1.31	-0.54	mA
G44,G45	III	VIII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
G46	III	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA

- See footnotes at end of table.

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Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G47	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G48	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G49	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G50	Iil	VIII	X	-2.05	0.09	-2.05	0.09	-2.05	0.09	mA
G51	Iil	VIII	X	-0.85	0.05	-0.85	0.05	-0.85	0.05	mA
G52	Iil	VIII	X	-2.28	0.1	-2.28	0.1	-2.28	0.1	mA
G53,G54	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G55	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G56	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G57-G59	Iil	VIII	X	-1.66	-0.54	-1.66	-0.54	-1.66	-0.54	mA
G60-G62	Iil	VIII	X	-1.26	-0.54	-1.26	-0.54	-1.26	-0.54	mA
H1-H28	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J28	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K4	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K5	Iozh	XI	X	-0.10	0.10	-0.10	0.10	-0.10	0.10	mA
K6-K9	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K10	Iozl	XI	X	-0.10	0.10	-0.10	0.10	-0.10	0.10	mA
L1-L3	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L4	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L5,L6	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L7,L8	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L9-L14	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L15	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L16,L17	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L18	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L19,L20	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L21	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L22,L23	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L24-L26	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	
L27,L28	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L29,L30	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L31-L34	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L35	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L36,L37	Vlc	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L38,L39	Vlc	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XIII	S	5.1	9.5	5.1	9.5	5.1	9.5	µF
M2	PSFC	XIII	S	7.8	14.6	7.8	14.6	7.8	14.6	µF
N	Icc	XIV	X	856	1284	856	1284	856	1284	mA
P1,P2	Functional	XV	X	----------		See table XV				
Q1,Q2	Dynamic	XVI	S	----------		See table XVI				
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI				
R1	Cin	XVIII	Q	4	100	4	100	4	100	pF
R2	Cin	XVIII	Q	6	125	6	125	6	125	pF
R3	Cin	XVIII	Q	2	50	2	50	2	50	pF
R4,R5	Cin	XVIII	Q	4	125	4	125	4	125	pF
R6,R7	Cin	XVIII	Q	2	75	2	75	2	75	pF
R8	Cin	XVIII	Q	4	75	4	75	4	75	pF
R9	Cin	XVIII	Q	4	100	4	100	4	100	pF
R10	Cin	XVIII	Q	2	75	2	75	2	75	pF
R11	Cin	XVIII	Q	8	125	8	125	8	125	pF
R12,R13	Cin	XVIII	Q	2	75	2	75	2	75	pF
R14	Cin	XVIII	Q	4	75	4	75	4	75	pF
R15	Cin	XVIII	Q	8	125	8	125	8	125	pF
R16	Cin	XVIII	Q	4	100	4	100	4	100	pF
R17	Cin	XVIII	Q	2	50	2	50	2	50	pF
R18,R19	Cin	XVIII	Q	2	75	2	75	2	75	pF
R20	Cin	XVIII	Q	2	50	2	50	2	50	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R21-R23	Cin	XVIII	Q	2	75	2	75	2	75	pF
R24,R25	Cin	XVIII	Q	4	75	4	75	4	75	pF
R26	Cin	XVIII	Q	2	75	2	75	2	75	pF
R27,R28	Cin	XVIII	Q	4	50	4	50	4	50	pF
R29	Cin	XVIII	Q	4	75	4	75	4	75	pF
R30	Cin	XVIII	Q	4	50	4	50	4	50	pF
R31	Cin	XVIII	Q	4	75	4	75	4	75	pF
R32	Cin	XVIII	Q	4	50	4	50	4	50	pF
R33	Cin	XVIII	Q	2	50	2	50	2	50	pF
R34	Cin	XVIII	Q	4	75	4	75	4	75	pF
R35,R36	Cin	XVIII	Q	2	100	2	100	2	100	pF
R37-R39	Cin	XVIII	Q	4	100	4	100	4	100	pF
R40	Cin	XVIII	Q	6	175	6	175	6	175	pF
R41,R42	Cin	XVIII	Q	6	125	6	125	6	125	pF
R43,R44	Cin	XVIII	Q	4	125	4	125	4	125	pF
R45	Cin	XVIII	Q	2	100	2	100	2	100	pF
R46	Cin	XVIII	Q	2	75	2	75	2	75	pF
R47	Cin	XVIII	Q	4	75	4	75	4	75	pF
R48	Cin	XVIII	Q	2	75	2	75	2	75	pF
R49	Cin	XVIII	Q	4	75	4	75	4	75	pF
R50	Cin	XVIII	Q	8	100	8	100	8	100	pF
R51	Cin	XVIII	Q	4	75	4	75	4	75	pF
R52	Cin	XVIII	Q	8	125	8	125	8	125	pF
R53,R54	Cin	XVIII	Q	2	50	2	50	2	50	pF
R55	Cin	XVIII	Q	2	75	2	75	2	75	pF
R56	Cin	XVIII	Q	2	100	2	100	2	100	pF
R57,R58	Cin	XVIII	Q	4	75	4	75	4	75	pF
R59	Cin	XVIII	Q	4	50	4	50	4	50	pF
R60-R62	Cin	XVIII	Q	4	75	4	75	4	75	pF
S1-S3	Input clamp diode latch-up	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond.	Parameter 5/	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
T1,T2	PSTA	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits	Notes
E1	CB MA CLR	7	0.0	Module shall pass all remaining tests.	1-4
E2	TP	9			
E3	NDRO BIT MOD	11			
E4	P BIT 06	14			
E5	P BIT 05	15			
E6	PRG FAULT FF	22			
E7	INTR PENDING	23			
E8	RST2 RNK FUL	24			
E9	IN IOC UPD	50			
E10	SET BUS REQ	55			
E11	ISTA WPBIT 0	57			
E12	ISTA WPBIT 1	61			
E13	ISTACK TO Y	63			
E14	P BIT 03	64			
E15	ISTA TO PIPE	65			
E16	P BIT 04	66			
E17	EWA PROTECT	69			
E18	AND CODE F	70			
E19	FLUSH ISTACK	71			
E20	IS RDPTR B1	79	0.0	Module shall pass all remaining tests.	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits	Notes
E21	SET ISTA EPT	82	0.0	Module shall pass all remaining tests.	1-4
E22	JMP CON TRUE	88			
E23	JMP CON TRUE	88			
E24	CP MA CLR	93			
E25	ISTACK FULL	96			
E26	CB MIC CTRL	97			
E27	PROG RUN FF	98			
E28	ISTA RPBIT 0	99	0.0	Module shall pass all remaining tests.	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E28
Power supply current limit	1.93 A	NA	E1-E28
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E28
Pattern voltage,			
high, (VH)	3.0 V	± 100 mV	E1-E28
low, (VL)	0.0 V	± 100 mV	E1-E28

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated test at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, no measurements are taken. Only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
			High level				
K1	P BIT 06	14	2.4	-0.05	0.05	mA	1-3
K2	P BIT 05	15					
K3	P BIT 03	64					
K4	P BIT 04	66		-0.05	0.05		
K5	JMP CON TRUE	88	2.4	-0.10	0.10	mA	1-3
			Low level				
K6	P BIT 06	14	0.4	-0.05	0.05	mA	1-3
K7	P BIT 05	15					
K8	P BIT 03	64					
K9	P BIT 04	66		-0.05	0.05		
K10	JMP CON TRUE	88	0.4	-0.10	0.10	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K10
Power supply current limit	1.93 A	NA	K1-K10
Forcing voltage, (Voh)	2.4 V	± 50 mV	K1-K5
(Vol)	0.4 V	± 50 mV	K6-K10

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage, high, (VH)	3.0 V	± 100 mV	K1-K10
low, (VL)	0.0 V	± 100 mV	K1-K10
Output current measurement	NA	± 5 μA	K1-K4
		± 10 μA	K5
		± 5 μA	K6-K9
	NA	± 10 μA	K10

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	MEM PROTECT	2	-18.0	-1.2	0.0	V	1-3
L2	CLII INTR EN	3					
L3	CPMEM RES ER	4		-1.2			
L4	WRITE ISTACK	8		-1.5			
L5	SEL ISTK SEL	12		-1.2			
L6	BREAKPOINT	13		-1.2			
L7	IR BIT 44	18		-1.5			
L8	IR BIT 06	20		-1.5			
L9	CK BRKPT STP	21		-1.2			
L10	CP CLOCK	26					
L11	MIC CTRL FMT	27					
L12	IR BIT 00	28					
L13	IR BIT 01	29					
L14	IR BIT 02	30		-1.2			
L15	IR BIT 41	32		-1.5			
L16	IR BIT 08	33		-1.2			
L17	IR BIT 10	34		-1.2			
L18	IR BIT 13	35		-1.5			
L19	IR BIT 15	36		-1.2			
L20	SGR-EQ TO IR	38		-1.2			
L21	STOIN ISTK D	46	-18.0	-1.5	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L22	CPSC ABUS EN	48	-18.0	-1.2	0.0	V	1-3
L23	JMAP OE	49		-1.2			
L24	ECW 13	52		-1.5			
L25	ECW 12	53					
L26	ECW 15	54		-1.5			
L27	IN ISTKW PTR	58		-1.2			
L28	CB CLK	59		-1.2			
L29	IR BIT 45	67		-1.5			
L30	CI INTR EN	68		-1.5			
L31	KILL PIPE	74		-1.2			
L32	CKINS PI REG	76					
L33	KIL PIPE	77					
L34	FORC I FETCH	78		-1.2			
L35	IN ISTKR PTR	80		-1.5			
L36	IR BIT 07	83		-1.2			
L37	IR BIT 09	84		-1.2			
L38	IR BIT 12	85		-1.5			
L39	IR BIT 14	86	-18.0	-1.5	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L39
Power supply current limit	NA	NA	L1-L39
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L39
Pattern voltage, high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L39 L1-L39
Input voltage measurement	NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1-L3 L4 L5,L6 L7,L8 L9-L14 L15 L16,L17 L18 L19,L20 L21 L22,L23 L24-L26 L27,L28 L29,L30 L31-L34 L35 L36,L37 L38,L39

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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